

Research Note: Event-Driven Mean Reversion in NIFTY 50

AlgoChowk Quant Engineer Assignment — Phase 1: Experimental Design

1. Formal Hypothesis

The core hypothesis under investigation is that extreme, short-term drawdowns in the equity index are driven primarily by rapid liquidity vacuums and panic selling rather than immediate fundamental deterioration. Therefore, the hypothesis states:

Following a daily close-to-close decline of 2.0% or greater in the NIFTY 50 index, the market exhibits a statistically significant mean-reverting recovery over the subsequent 5 trading days, generating a net positive expected value that outperforms the unconditional baseline return.

2. Event & Recovery Definitions

To ensure the experiment is strictly measurable and reproducible, qualitative terms are defined mathematically:

- **Significant Fall (The Event):** A daily return $R_t \leq -2.0\%$, calculated as $(C_t - C_{t-1}) / C_{t-1}$, where C represents the official closing price. A static absolute threshold is preferred over a rolling standard deviation to capture systemic panic rather than standard local volatility.
- **Recovery:** The cumulative percentage return generated during the designated holding period, explicitly net of estimated transaction costs.

3. Entry & Exit Mechanics

A frequent error in retail backtesting is the introduction of look-ahead bias by entering trades at the close of the event day. To model realistic market microstructure:

- **Entry Execution:** Entry occurs exactly at the Open price of day $T+1$ (O_{t+1}). The official closing price of day T is unknown until continuous trading halts; therefore, executing an automated entry order at C_t is practically impossible without severe slippage.
- **Exit Execution:** Exit occurs at the Close price at the end of the predefined holding period.

4. Holding Period

The holding period is strictly defined as **5 trading days**. The exit execution occurs at the close of day $T+5$. A 5-day window is hypothesized to be optimal because it provides sufficient duration for short-term liquidity providers to stabilize the market and for panic-sellers to exhaust their inventory, while remaining short enough to avoid exposure to longer-term macroeconomic shifts that are untethered from the initial drop catalyst.

5. Test Period & Data Architecture

The research engine relies on historical daily OHLC data. The parameters are as follows:

- **Data Source:** Yahoo Finance (Ticker: ^NSEI). This is a widely credible and easily reproducible dataset.
- **Global Date Range:** January 1, 2010, to August 31, 2026. This extensive period encompasses multiple distinct market regimes, including the 2011 European debt crisis fallout, the 2020 COVID-19 high-volatility crash, and subsequent low-volatility bull runs.
- **Validation Split:** To mitigate the risk of data snooping and post-hoc parameter fitting, the data is chronologically partitioned:
 - **Research / Development (In-Sample):** Jan 1, 2010 – Dec 31, 2021.
 - **Out-of-Sample Validation:** Jan 1, 2022 – Aug 31, 2026.

6. Transaction Costs (Friction)

Gross returns are mathematically meaningless without accounting for execution friction. A total transaction cost of **20 basis points (0.20%)** per round-trip trade is applied. This assumption incorporates 10 bps on entry and 10 bps on exit to accurately account for retail or tier-2 institutional brokerage fees, statutory levies (like STT in India), and slippage resulting from executing market-on-open orders during high-volatility sessions.

7. Key Assumptions & State Management

The most critical engineering assumption involves the handling of overlapping event triggers. High-volatility market crashes often cluster (e.g., March 2020), meaning multiple $\leq -2.0\%$ days can occur within a single 5-day holding period.

Event Independence: Treating closely clustered drops as independent observations artificially inflates statistical significance and assumes unlimited capital allocation. Therefore, a strict state-management protocol is enforced: If a qualifying event triggers an entry, the system enters an "active" state. Any subsequent $\leq -2.0\%$ drops that occur while the $T+5$ holding period is active are explicitly ignored. The system must completely exit the current trade before a new event can trigger an entry. This guarantees that every measured trade in the sample is a fully independent observation.

Objective Summary: The design outlined above prioritizes falsifiability over finding the highest historical return. By explicitly removing look-ahead bias, applying realistic transaction costs, enforcing strict out-of-sample partitioning, and ensuring statistical independence of events, the subsequent code implementation will yield an honest assessment of whether the mean-reversion hypothesis deserves to be believed.